AKM Shahariar Azad Rabby

Birmingham, Alabama | shahariar@rabby.dev | rabby.dev | Google Scholar | GitHub

Academic Profile

I develop computer-vision methods for segmenting thin structures, with a focus on structural reliability in cracks and pulmonary fissures. I also evaluate such methods on retinal vessels and study why good Dice or F1 scores can mask broken connections and missing branches. Earlier work in Bangla OCR and deployed vision-language systems informs my research on multimodal evidence and efficient inference.

Scholarly record: 50 accepted or published peer-reviewed papers; 16 first-author papers; over 1,000 citations; h-index 17; i10-index 25.

Mentored 13 undergraduate researchers to first-author peer-reviewed publications.

Research interests: Trustworthy and efficient computer vision; structural reliability and thin-structure segmentation; multimodal evidence evaluation; adaptive and deployable AI; medical imaging and document intelligence.

Education

Ph.D. in Computer Science, University of Alabama at Birmingham Aug 2021–Apr 2026

Dissertation: Precise Segmentation of Thin, Elongated, and Underrepresented Structures. Advisor: Dr. Chengcui Zhang.

M.S. in Computer Science, University of Alabama at Birmingham Aug 2021–Apr 2023

Graduate coursework: Machine Learning, Deep Learning, Data Science, Cloud Computing, Algorithms.

B.S. in Computer Science and Engineering, Daffodil International University Jan 2015–Dec 2018

Thesis: Ekush: Bangla Handwritten Character Data Repository for NLP Research.

Academic and Research Appointments

Research Scientist, University of Alabama at Birmingham Jun 2026–Present
Center for Lung Analytics & Imaging Research (CLAIR) / Knowledge Discovery & Data Mining Lab

- Research on CT-based COPD detection, pulmonary fissure segmentation, and 3D fissure continuity and incompleteness analysis within NIH/NHLBI-supported imaging studies; ongoing development of a quantitative fissure-integrity score.
- Develop 3D CT processing and GPU inference workflows for multicenter evaluation and medical-imaging research.

Graduate Research Assistant, University of Alabama at Birmingham Aug 2021–May 2026
CLAIR / Knowledge Discovery & Data Mining Lab

- Led COPDxNet architecture and multicenter evaluation; designed STRIPE; developed thin-structure segmentation methods and benchmark datasets.
- Contributed image registration and voting-target localization to NSF-funded ballot-image research and built inference workflows with Triton Inference Server.

Selected Research Contributions

STRIPE — directional segmentation (accepted, IEEE BIBM 2026): Designed an approximately 5.1M-parameter directional encoder–decoder for pulmonary fissures; Dice 0.873 and median HD95 0.72 mm in COPDGene five-fold cross-validation. Proceedings forthcoming.

dCrack / dCrack61K — edge-aware segmentation (IEEE IRI 2025): Developed Edge-Guided Attention and led construction of dCrack61K (61,943 images from 36 sources, over 2,000 corrected annotations). The dCrack++ architecture achieved F1 0.8251 on CHASE_DB1 retinal vessels using models trained from scratch under five-fold cross-validation.

COPDxNet — multicenter CT-based COPD classification (under review): Led architecture development and multicenter evaluation of a 10.6M-parameter 3D CNN using 13,043 COPDGene CT scans from 21 centers, with participant-level train/test separation. Standard-dose AUC 0.92 and low-dose AUC 0.88; external AUC 0.92 (SPIROMICS) and 0.82 (NLST).

THREAD — efficient continuity-aware thin-structure vision (under review): Developed a crack-segmentation architecture combining directional feature extraction and structural refinement. Across four benchmarks, mean F1@0.5 was 0.8250; on Crack500, F1@0.5 was 0.7481 and clDice was 0.8202. The model has 5.634M parameters and requires 9.326G FLOPs (48.6% below SCSegamba's 18.157G). Inference reached 75.35 FPS on NVIDIA A6000 and 4.711 FPS at 256×256 or 0.970 FPS at 512×512 on Raspberry Pi 5. The manuscript includes cross-dataset zero-shot evaluation.

Low-resource document AI: Developed Bangla handwriting/OCR datasets and systems, including Ekush (673,000 images; 242 classes), and contributed to EMNLP 2023 and EACL 2026 Industry Track papers.

Teaching and Research Mentoring

Course Instructor, AIM 646: LLM Development in Medicine, UAB Heersink School of Medicine Fall 2026
Three-credit graduate course; in-person and online instruction

- Teach graduate-level language-model development through implementation, retrieval-augmented applications, and responsible, evidence-based evaluation in medical-domain settings.

Lecturer (Part-Time), Daffodil International University Jan 2019–Dec 2020
Department of Computer Science and Engineering

- Taught CSE 412 Artificial Intelligence and CSE 413L AI Lab to 50+ students per semester; used Python/PyTorch implementations, evaluation, debugging, and applied AI projects.

Research mentorship: Mentored 13 undergraduate researchers to first-author peer-reviewed publications. Research Supervisor, DIU NLP & ML Lab (2019–2021); currently International Advisor.

Industry and Technical Leadership

Senior ML Engineer / Technical Lead (Part-Time), Apurba Technologies Nov 2025–Present
Remote collaboration from Birmingham, Alabama

- Deployed Qwen2.5-VL inference with vLLM on one NVIDIA L40S GPU, using LoRA, FP8, and Flash Attention 2, for Kagoj.ai document-AI workloads.
- Contributed to the Kagoj.ai platform, serving approximately 30,000 daily users. Platform-wide peaks reached 1,000–2,000 API requests per second, with recorded end-to-end API response latencies of 84 ms (p50) and 91 ms (p95).
- Led a 13-engineer team migrating AI workloads to Kafka-based event-driven microservices for document processing.
- Architected a privacy-conscious, PHI-aware retrieval-augmented system for medical-bill auditing; contributed to industry research collaborations in OCR/document intelligence.

Machine Learning Engineer → Team Lead, Apurba Technologies Oct 2019–Aug 2021
Dhaka, Bangladesh

- Built document-digitization ML pipelines; optimized Triton/ONNX inference for 18.6× speedup and 10× memory reduction; grew team from 3 to 10+ and scaled services to 10,000+ daily requests with 99.5% uptime.

Artificial Intelligence Engineer (Contract), Genex Digital Apr 2019–Oct 2019
Dhaka, Bangladesh

- Developed multilingual Bangla, English, and Banglish conversational AI for telecom and banking applications.

Sponsored Research Participation

NIH/NHLBI R01 HL151421 — "Structural Determinants of Disease Progression in COPD" (2022–Present): PIs: Surya Bhatt, MD, and Arie Nakhmani, PhD. Research Scientist (June 2026–present); Graduate Research Assistant (2022–May 2026). Contributions: COPDxNet, STRIPE, multicenter CT analysis, and scalable imaging AI.

NSF Bubble Aid — "Assistive AI to Improve the Robustness and Security of Reading Hand-Marked Ballots" (2022–2027): Multi-university collaborative project (award 2154589; UAB award CNS-2154507). Dr. Chengcui Zhang, UAB investigator. Graduate Research Assistant, first project year; image registration, voting-target localization, and IEEE IRI 2023 collaboration.

Awards and Fellowships

- CS Outstanding Ph.D. Student Award, UAB (2026).
- Blazer Graduate Research Fellowship, UAB (2021–2022); UAB Graduate Dean's Scholarship (2022).
- Best Paper Award, RTIP2R (2018); Best Paper Award, IEMIS (2018).
- CVPR Travel Grant (2022); Barrett R. and Oceana A. Bryant Endowed Awards (2022, 2024).
- Best Student Research Award, DIU CSE (2018, 2019); First Runner-Up, SBCL 3-Minute Research Pitch Competition (2022).

Academic Service and Leadership

- Workshop Organizer, Workshop on Vision-Based Understanding for Low-Resource Languages (WVLL), IEEE/CVF WACV 2024, Waikoloa, Hawaii.
- Organizer (upcoming), Embodied AI Agents, Robotics, and IoT Special Track, FLAIRS-40 (2027), St. Pete Beach, Florida (scheduled May 24–27, 2027).
- Completed 33+ peer reviews for journal and conference venues, including IEEE TDSC, IEEE TMM, IJMDEM, ACM Multimedia, WACV, ICMLA, AAAI, and CVPR; maintains a Web of Science/Publons reviewer record.
- International Advisor, DIU NLP & ML Lab; previously Research Supervisor (2019–2021). Webmaster, UAB Graduate Student Government (2024–2025).

Research Artifacts

- Ekush: 673,000 Bangla handwritten character images in 242 classes; dCrack61K: 61,943 images from 36 sources, with 2,000+ corrected annotations; ISHARA-LIPI: 1,800 images across 36 classes.

Technical Expertise

Methods: Computer vision, thin-structure segmentation, deep learning, medical CT analysis, evaluation under domain shift, retrieval-augmented and multimodal AI.

Frameworks: PyTorch, TensorFlow, MONAI, Transformers, timm, vLLM.

Systems: Triton Inference Server, ONNX Runtime, TensorRT, Docker, FastAPI, Kafka, AWS, GPU inference, quantization.

Programming and tools: Python, C++, SQL, Bash, LaTeX; LoRA, FP8, Flash Attention 2.

Professional memberships: IEEE (2021–present); American Thoracic Society (2023–present); Bangladesh Society for Private University Academics (2022–present).

Selected Publications (9 of 50)

J1. Saha, P., Chaudhary, M.F.A., **Rabby, A.S.A.**, Bodduluri, S., Nakhmani, A., Zhang, C., Bhatt, S.P. "Machine Learning Based Detection of True Ventilatory Restriction." Respiratory Medicine (Elsevier), 2025.

C1. **Rabby, A.S.A.**, Zhang, C., Nakhmani, A., Bodduluri, S., Bhatt, S.P. "STRIPE: Efficient Directional Feature Encoding for Ultra-Thin Pulmonary Fissure Segmentation." Accepted at IEEE BIBM 2026 (proceedings forthcoming).

C3. **Rabby, A.S.A.**, Zhang, C. "dCrack: Enhancing Fine-Grained Crack Segmentation with Edge-Guided Attention." IEEE International Conference on Information Reuse and Integration (IRI), 2025.

C4. **Rabby, A.S.A.**, Saha, P., Abujar, S., Rahman, F. "One Model Multiple Insights: CNNs for Medical Image Classification." IEEE International Conference on Big Data, pp. 3935–3941, 2024. DOI: 10.1109/BigData62323.2024.10825607.

C9. **Rabby, A.S.A.**, Islam, M.M., Islam, Z., Hasan, N., Rahman, F. "Towards Building a Robust Large-Scale Bangla Text Recognition Solution Using a Unique Multiple-Domain Character-Based Document Recognition Approach." IEEE International Conference on Machine Learning and Applications (ICMLA), pp. 1393–1399, 2021. DOI: 10.1109/ICMLA52953.2021.00225.

IT1. Islam, S.M.J., Hasan, M.M., Ovi, M.H., **Rabby, A.S.A.**, Rahman, F. "BornoDrishti: Leveraging Vision Encoders and Domain-Adaptive Learning for Bangla OCR on Diverse Documents." EACL Industry Track, pp. 278–286, 2026.

IT2. Ali, H., **Rabby, A.S.A.**, Islam, M.M., Mahamud, A.K.M., Hasan, N., Rahman, F. "Gold Standard Bangla OCR Dataset: An In-Depth Look at Data Preprocessing and Annotation Processes." EMNLP Industry Track, 2023.

W1. **Rabby, A.S.A.**, Ali, H., Islam, M.M., Abujar, S., Rahman, F. "Enhancement of Bengali OCR by Specialized Models and Advanced Techniques for Diverse Document Types." IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops, 2024.

C34. **Rabby, A.S.A.**, Haque, S., Islam, M.S., Abujar, S., Hossain, S.A. "Ekush: A Multipurpose and Multitype Comprehensive Database for Online Off-Line Bangla Handwritten Characters." International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R), Springer, 2018. [Best Paper Award]

Publications

50 accepted or published peer-reviewed papers: 1 journal article, 45 conference papers, 2 industry-track papers, and 2 workshop papers. The 16 first-author papers are included in this total. Preprints, under-review manuscripts, and conference abstracts appear separately.

Journal Article

J1. Saha, P., Chaudhary, M.F.A., **Rabby, A.S.A.**, Bodduluri, S., Nakhmani, A., Zhang, C., Bhatt, S.P. "Machine Learning Based Detection of True Ventilatory Restriction." Respiratory Medicine (Elsevier), 2025.

Peer-Reviewed Conference Papers

C1. **Rabby, A.S.A.**, Zhang, C., Nakhmani, A., Bodduluri, S., Bhatt, S.P. "STRIPE: Efficient Directional Feature Encoding for Ultra-Thin Pulmonary Fissure Segmentation." Accepted at IEEE BIBM 2026 (proceedings forthcoming).

C2. Hasan, M.M., Islam, S.M.J., **Rabby, A.S.A.**, Rahman, F. "Beyond Characters: Position-Aware Metric and a Lightweight LLM for Low-Resource Bangla Punctuation Restoration." IEEE International Conference on Big Data (Big Data), 2025.

C3. **Rabby, A.S.A.**, Zhang, C. "dCrack: Enhancing Fine-Grained Crack Segmentation with Edge-Guided Attention." IEEE International Conference on Information Reuse and Integration (IRI), 2025.

C4. **Rabby, A.S.A.**, Saha, P., Abujar, S., Rahman, F. "One Model Multiple Insights: CNNs for Medical Image Classification." IEEE International Conference on Big Data, pp. 3935–3941, 2024. DOI: 10.1109/BigData62323.2024.10825607.

C5. Akanda, M., Ahmed, M., **Rabby, A.S.A.**, Rahman, F. "Optimum Deep Learning Method for Document Layout Analysis in Low Resource Languages." ACM Southeast Conference, 2024.

C6. **Rabby, A.S.A.**, Ali, H., Islam, M.M., Rahman, F. "Versatile Bengali OCR: Document Analysis Technique for Varied Document Styles and Content." IEEE International Conference on Big Data, pp. 1965–1969, 2023. DOI: 10.1109/BigData59044.2023.10386582.

C7. Zhao, F., Zhang, C., Saxena, N., Wallach, D., **Rabby, A.S.A.** "Ballot Tabulation Using Deep Learning." IEEE International Conference on Information Reuse and Integration (IRI), 2023.

C8. Das, A., **Rabby, A.S.A.**, Kowsar, I., Rahman, F. "A Deep Learning-based Unified Solution for Character Recognition." International Conference on Pattern Recognition (ICPR), 2022.

C9. **Rabby, A.S.A.**, Islam, M.M., Islam, Z., Hasan, N., Rahman, F. "Towards Building a Robust Large-Scale Bangla Text Recognition Solution Using a Unique Multiple-Domain Character-Based Document Recognition Approach." IEEE International Conference on Machine Learning and Applications (ICMLA), pp. 1393–1399, 2021. DOI: 10.1109/ICMLA52953.2021.00225.

C10. Rahman, F., Rahman, A., **Rabby, A.S.A.**, Rifat, M.J.R., Banik, M., Islam, M.M., Islam, A., Aziz, N.A., Meyer, R., Kriak, J., Goldblatt, S. "Modeling Influenza with a Forest Deep Neural Network Utilizing a Virtualized Clinical Semantic Network." IEEE International Conference on Big Data, pp. 4753–4760, 2021. DOI: 10.1109/BigData52589.2021.9671507.

C11. Islam, M.M., Das, A., Kowsar, I., **Rabby, A.S.A.**, Hasan, N., Rahman, F. "Towards building a Bangla text recognition solution with a Multi-Headed CNN architecture." IEEE International Conference on Big Data, pp. 1061–1067, 2021. DOI: 10.1109/BigData52589.2021.9671653.

C12. Rahman, F., Rahman, A., **Rabby, A.S.A.**, Rifat, M.J.R., Banik, M., Islam, M., Islam, A., Aziz, N.A., Meyer, R., Kriak, J., Goldblatt, S. "Disease Modeling with a Forest Deep Neural Network Utilizing NLP and a Virtualized Clinical Semantic Network." IEEE International Conference on Tools with Artificial Intelligence (ICTAI), 2021. DOI: 10.1109/ICTAI52525.2021.00150.

C13. Islam, M.S., Khushbu, S.A., **Rabby, A.S.A.**, Bhuiyan, T. "A Study on Dengue Fever in Bangladesh: Predicting the Probability of Dengue Infection with External Behavior." IEEE International Conference on Intelligent Computing and Control Systems (ICICCS), 2021.

C14. Ferdous, J., Karmaker, S., **Rabby, A.S.A.**, Hossain, S.A. "MatriVasha: A Multipurpose Comprehensive Database for Bangla Handwritten Compound Characters." Emerging Technologies in Data Mining and Information Security (IEMIS), Springer, 2021.

C15. Sany, M.H., Keya, M., Khushbu, S.A., **Rabby, A.S.A.**, Masum, A.K.M. "An Opinion Mining of Text in COVID-19 Issues Along with Comparative Study in ML, BERT & RNN." International Conference on Deep Learning, Artificial Intelligence and Robotics, Springer, 2021.

C16. Islam, M.M., **Rabby, A.S.A.**, Hasan, N., Nahar, J., Rahman, F. "A Novel Bangla Font Recognition Approach Using Deep Learning." IEMIS, Springer, 2021.

C17. **Rabby, A.S.A.**, Islam, M.M., Hasan, N., Nahar, J., Rahman, F. "A Novel Deep Learning Character-Level Solution to Detect Language and Printing Style from a Bilingual Scanned Document." IEEE International Conference on Big Data (Big Data), 2020.

C18. **Rabby, A.S.A.**, Islam, M.M., Hasan, N., Nahar, J., Rahman, F. "Borno: Bangla Handwritten Character Recognition Using a Multiclass Convolutional Neural Network." Future Technologies Conference (FTC), Springer, 2020.

C19. **Rabby, A.S.A.**, Islam, A., Rahman, F. "Establishing a Formal Benchmarking Process for Sentiment Analysis for the Bangla Language." Future Technologies Conference (FTC), Springer, 2020.

C20. **Rabby, A.S.A.**, Islam, M.M., Hasan, N., Nahar, J., Rahman, F. "Language Detection using Convolutional Neural Network." IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2020.

C21. Roy, K., Hasan, M., Fuhad, K.M.F., Mohammed, N., **Rabby, A.S.A.**, Hasan, N., Nahar, J., Rahman, F. "Bangla Part of Speech Tagging Using Contextual Embeddings and Oversampling Techniques." Future Technologies Conference (FTC), Springer, 2020.

C22. Jamal, S., Bappy, T.H., **Rabby, A.S.A.** "Brazilian Forest Fire Analysis: An Unsupervised Approach." Soft Computing Techniques and Applications, Springer, 2020.

C23. Jamal, S., Bappy, T.H., Pervin, R., **Rabby, A.S.A.** "Weather Status Prediction of Dhaka City Using Machine Learning." Computational Methods and Data Engineering, Springer, 2020.

C24. Islam, A., Rahman, F., **Rabby, A.S.A.** "Sankhya: An Unbiased Benchmark for Bangla Handwritten Digits Recognition." IEEE International Conference on Big Data (Big Data), 2019.

C25. Hasan, H.M.R., **Rabby, A.S.A.**, Islam, M.T., Hossain, S.A. "Machine Learning Algorithm for Student's Performance Prediction." IEEE ICCCNT, 2019.

C26. **Rabby, A.S.A.**, Mamata, R., Laboni, M.A., Ohidujjaman, Abujar, S. "Machine Learning Applied to Kidney Disease Prediction: Comparison Study." IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2019. DOI: 10.1109/ICCCNT45670.2019.8944799.

C27. Islam, M.M., **Rabby, A.S.A.**, Arfin, M.H.R., Hossain, S.A. "PataNET: A Convolutional Neural Networks to Identify Plant from Leaf Images." IEEE ICCCNT, 2019.

C28. Sheikh, M.H., Mim, T.T., Reza, M.S., **Rabby, A.S.A.**, Hossain, S.A. "Detection of Maize and Peach Leaf Diseases Using Image Processing." IEEE ICCCNT, 2019.

C29. Masum, A.K.M., Abujar, S., Talukder, M.A.I., **Rabby, A.S.A.**, Hossain, S.A. "Abstractive Method of Text Summarization with Sequence to Sequence RNNs." IEEE ICCCNT, 2019.

C30. Khan, T.A., Islam, M.S., Ullah, S.M., **Rabby, A.S.A.** "A Machine Learning Approach to Recognize Junk Food." IEEE ICCCNT, 2019.

C31. Arif, M.A.I., Sany, S.I., Nahin, F.I., **Rabby, A.S.A.** "Comparison Study: Product Demand Forecasting with Machine Learning for Shop." IEEE International Conference System Modeling and Advancement in Research Trends (SMART), 2019.

C32. Al Mamun, M.A., Kadir, I., **Rabby, A.S.A.**, Al Azmi, A. "Bangla Music Genre Classification Using Neural Network." IEEE SMART, 2019.

C33. Abdullha, A., Habib, Y., Masum, M.R.I., **Rabby, A.S.A.** "Countries Condition of Forestation and Trees Percentage using Machine Learning." IEEE SMART, 2019.

C34. **Rabby, A.S.A.**, Haque, S., Islam, M.S., Abujar, S., Hossain, S.A. "Ekush: A Multipurpose and Multitype Comprehensive Database for Online Off-Line Bangla Handwritten Characters." International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R), Springer, 2018. [Best Paper Award]

C35. Islam, M.S., Khushbu, S.A., **Rabby, A.S.A.**, Hossain, S.A. "Ishara-Lipi: The First Complete Multipurpose Dataset for Bangla Sign Language." International Conference on Bangla Speech and Language Processing (ICBSLP), IEEE, 2018.

C36. Islam, M.S., Mousumi, S.S.S., Jessan, N.A., **Rabby, A.S.A.**, Abujar, S., Hossain, S.A. "Ishara-Bochon: The First Multipurpose Open Access Dataset for Bangla Sign Language Isolated Digits." RTIP2R, Springer, 2018.

C37. Haque, S., **Rabby, A.S.A.**, Laboni, M.A., Neehal, N., Hossain, S.A. "ExNET: Deep Neural Network for Exercise Pose Detection." RTIP2R, Springer, 2018.

C38. Haque, S., **Rabby, A.S.A.**, Islam, M.S., Hossain, S.A. "ShonkhaNet: A Dynamic Routing for Bangla Handwritten Digit Recognition Using Capsule Network." RTIP2R, Springer, 2018.

C39. Haque, S., Shahinoor, S.A., **Rabby, A.S.A.**, Abujar, S., Hossain, S.A. "OnkoGan: Bangla Handwritten Digit Generation with Deep Convolutional Generative Adversarial Networks." RTIP2R, Springer, 2018.

C40. Islam, M.S., Mousumi, S.S.S., **Rabby, A.S.A.**, Hossain, S.A. "A Simple and Mighty Arrowhead Detection Technique of Bangla Sign Language Characters with CNN." RTIP2R, Springer, 2018.

C41. **Rabby, A.S.A.**, Abujar, S., Haque, S., Hossain, S.A. "Bangla Handwritten Digit Recognition Using Convolutional Neural Network." Emerging Technologies in Data Mining and Information Security, Springer, 2018.

C42. **Rabby, A.S.A.**, Haque, S., Abujar, S., Hossain, S.A. "EkushNet: Using Convolutional Neural Network for Bangla Handwritten Recognition." Procedia Computer Science (Elsevier), Vol. 143, pp. 603-610, 2018.

C43. Islam, S., Mousumi, S.S.S., **Rabby, A.S.A.**, Hossain, S.A., Abujar, S. "A Potent Model to Recognize Bangla Sign Language Digits Using Convolutional Neural Network." Procedia Computer Science (Elsevier), Vol. 143, pp. 611-618, 2018.

C44. **Rabby, A.S.A.**, Haque, S., Islam, S., Abujar, S., Hossain, S.A. "BornoNet: Bangla Handwritten Characters Recognition Using Convolutional Neural Network." Procedia Computer Science (Elsevier), Vol. 143, pp. 528-535, 2018.

C45. **Rabby, A.S.A.**, Haque, S., Shahinoor, S.A., Abujar, S., Hossain, S.A. "A Universal Way to Collect and Process Handwritten Data for Any Language." Procedia Computer Science (Elsevier), Vol. 143, pp. 502-509, 2018.

Industry-Track Papers

IT1. Islam, S.M.J., Hasan, M.M., Ovi, M.H., **Rabby, A.S.A.**, Rahman, F. "BornoDrishti: Leveraging Vision Encoders and Domain-Adaptive Learning for Bangla OCR on Diverse Documents." EACL Industry Track, pp. 278–286, 2026.

IT2. Ali, H., **Rabby, A.S.A.**, Islam, M.M., Mahamud, A.K.M., Hasan, N., Rahman, F. "Gold Standard Bangla OCR Dataset: An In-Depth Look at Data Preprocessing and Annotation Processes." EMNLP Industry Track, 2023.

Workshop Papers

W1. **Rabby, A.S.A.**, Ali, H., Islam, M.M., Abujar, S., Rahman, F. "Enhancement of Bengali OCR by Specialized Models and Advanced Techniques for Diverse Document Types." IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops, 2024.

W2. Mollah, S., Rakib, M., Mohammed, N., Wasek, M., **Rabby, A.S.A.**, Rahman, F. "Automated Mapping of Healthcare Concepts to a Standardized Healthcare Taxonomy." Tiny Papers @ ICLR, 2023.

Manuscripts Under Review and Preprints

P1. **Rabby, A.S.A.**, Chaudhary, M.F.A., Saha, P., Sthanam, V., Nakhmani, A., Zhang, C., Barr, R.G., Bon, J., Cooper, C.B., Curtis, J.L., Hoffman, E.A., Paine III, R., Puliyakote, A.K., Schroeder, J.D., Sieren, J.C., Smith, B.M., Woodruff, P.G., Reinhardt, J.M., Bhatt, S.P., Bodduluri, S. "Light Convolutional Neural Network to Detect Chronic Obstructive Pulmonary Disease (COPDxNet): A Multicenter Model Development and External Validation Study." medRxiv, 2025. Under review at The Lancet Digital Health.

P2. **Rabby, A.S.A.**, Zhang, C. "THREAD: Efficient Directional Context and Structural Refinement for Crack Segmentation." Manuscript under review, 2026.

P3. **Rabby, A.S.A.**, Zhang, C. "BeyondPixels: A Comprehensive Review of the Evolution of Neural Radiance Fields." arXiv preprint arXiv:2306.03000, 2023.

Conference Abstracts and Presentations

A1. **Rabby, A.S.A.**, Saha, P., Nakhmani, A., Reinhardt, J.M., Zhang, C., Bhatt, S.P., Bodduluri, S. "COPDxNet: An End-to-End Deep Neural Network for COPD Detection." American Thoracic Society (ATS) International Conference, 2024.